Reverse Video and a Faster Serial Port

For Osborne 1 owners: how to change your serial port baud rate, and how to change your underline cursor to a reversed-video one.

Thom Hogan

Warning
This article describes how to modify your Osborne 1. If you have a warranty still in effect (such as one through the Xerox® Americare® program or other such service), you should not perform the modification without becoming fully aware of the consequences.

While Osborne Computer Corporation may be gone, about 150,000 Osborne 1 computers aren't, and their users have no direct factory support and only scant information concerning the workings of the machine.

This article addresses two aspects of the Osborne 1 that many owners wish could be changed: the way in which the cursor displays, and faster baud rates for the serial port. This information, while developed internally by the company, was not promoted by Osborne Computer Corporation for a number of reasons, the primary of which was the problem of warranty repair on modified machines.

Don't attempt these modifications yourself if you are not qualified. You will be making cuts, adding jumpers, and soldering directly on integrated circuits if you follow these instructions. A slip of the soldering iron can be fatal (at least to the Osborne 1). If you want the modifications but haven't worked with delicate electronic components before, see if your local computer dealer or someone in your local Osborne user's group is willing to make them for you. Expertly done, the modifications shouldn't take more thanan hour to make, and the parts cost is minimal — so $50-$100 is a reasonable fee for the work.

The end result of the modifications will be a computer that has an easily-recognizable cursor and is capable of communicating at as fast as 9600 baud.

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Reversing the Cursor

During my short tenure at Osborne Computer Corporation (1981-1982) as manager of the software and publications departments and as a key member of the new product design council at the company, I had argued that the underline cursor on the Osborne 1 was not appropriate, especially considering the small screen — you just couldn't find it fast enough during the word processing. I discussed this problem with the computer's designer, Lee Felsenstein, who informed me that at one time the Osborne 1 computers had reverse video, but that the cursor was switched to an underline before production.

To make a long story short, I had Lee rework my Osborne 1, and started a

Reverse Video Schematic Change

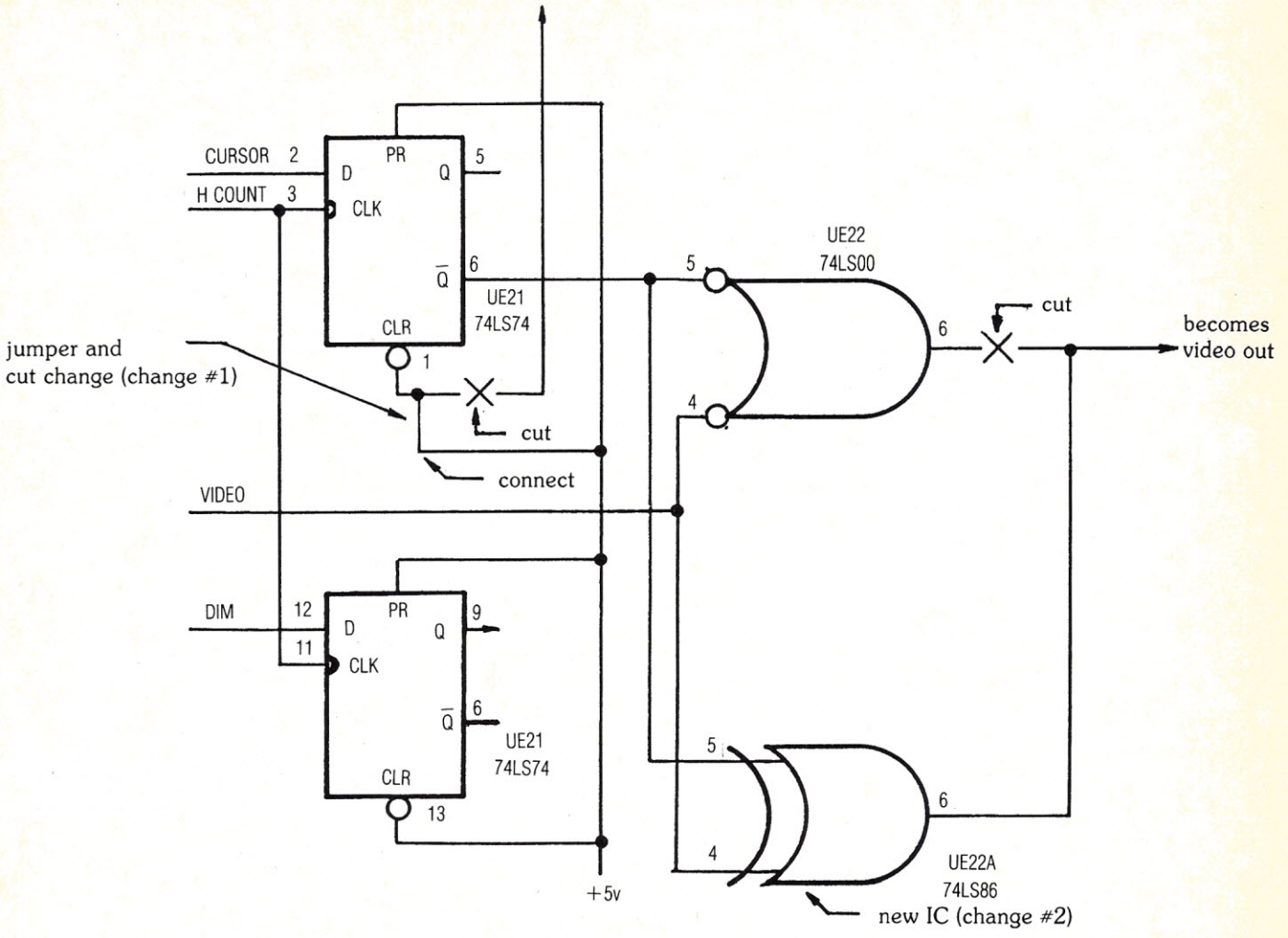


Figure 1. *A schematic of the reverse video cursor modification and how it affects the original circuitry.*

long vigil of trying to convince others in the company to make this permanent change. No luck. How about publishing the modifications? No luck there either, as the company was worried about the legal implications of publishing modifications.

The modification for reverse video cursor is actually quite simple. Currently, the video goes through one portion of a 74LS00 logic gate just prior to heading towards the video display. Simply switching the type of logic gate used, and making one small cut on the electronics board gives a reverse video cursor. Figure 1 shows a schematic of the modification and how it affects the original circuitry.

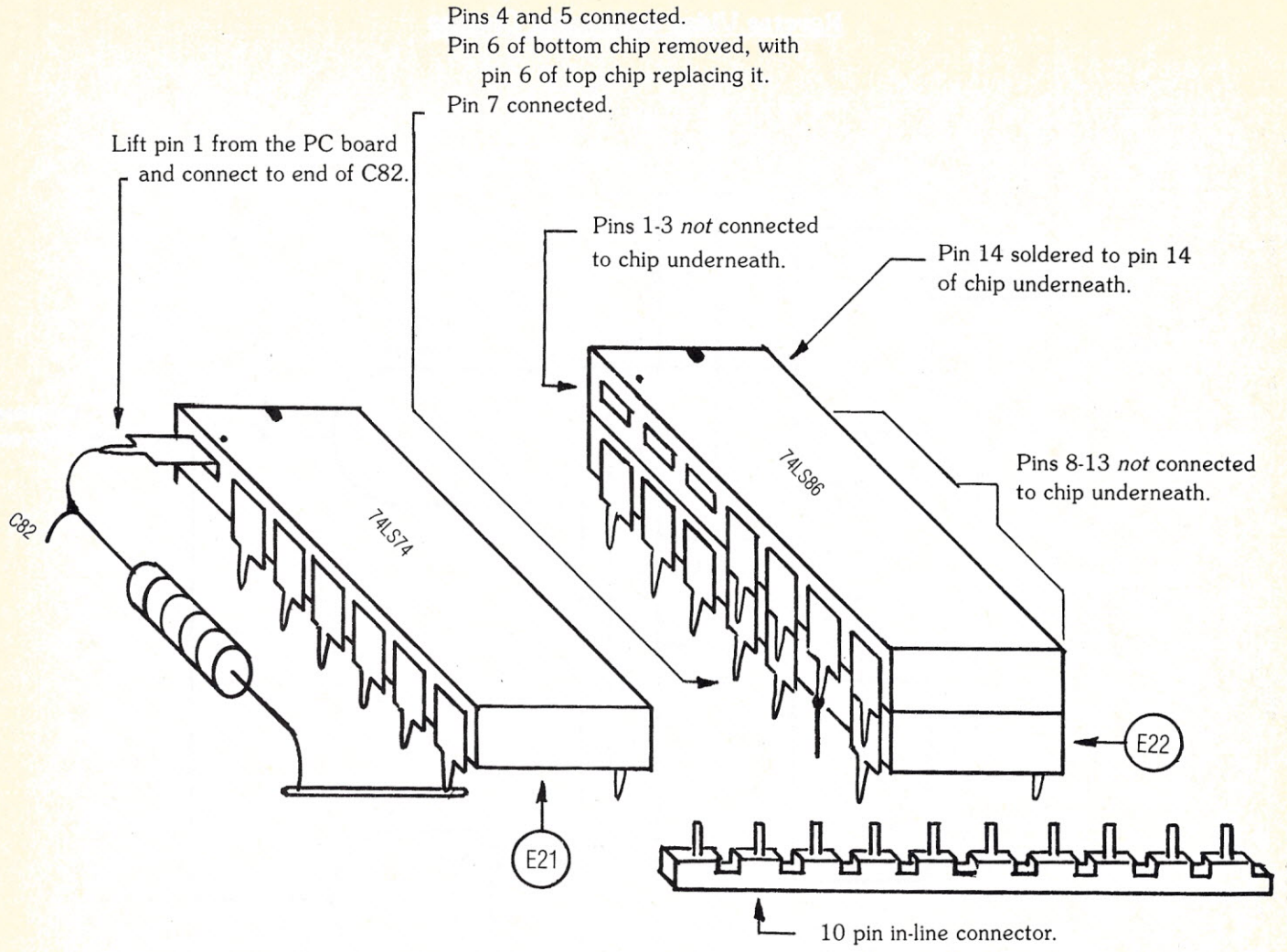
Logical considerations aside, the practical problems of changing the gate used to implement the cursor became important — we don't want to have to unsolder an IC, since that's not an easy process. Besides, the rest of the 74LS00 is needed for some other functions. Instead, we'll simply piggyback a new IC on top of the old one, and make our connections to it. Such a practice is relatively common for minor changes such as we are making.

First, you'll have to open up the Osborne 1 case and remove the main electronics board. If you don't already know how to do this, you shouldn't be making this modification! Opening the Osborne 1 is a simple process. Make sure that you note the orientation of the various cables as you unplug them.

Now we're ready for the delicate work. All you need is two short lengths of wire-wrap or other suitable wire and a 74LS86 to perform the change. Let's work in an ordered, logical fashion:

1. Take the 74LS86 chip and carefully cut off pins 1-3 and 8-13 as close to the plastic housing as possible. We aren't going to use these pins.

2. While you've got the snips out, very carefully cut off pin 6 of IC UE22, the 74LS00 located at position E22 on the main circuit board.



NOTE: ICs shown are located at board positions E21 and E22.

Figure 2. *The modifications to the ICs located at board positions E21 and E22 to produce a reverse video cursor.*

Make sure that you don't damage the solder pad that pin 6 connects to or the other pins on the 74LS00.

3. Next, carefully lift pin 1 of UE21, the 74LS74 located at position E21 on the main circuit board.

4. Place the 74LS86 on top of the 74LS00 at E22. You're going to solder pins 4, 5, 7, and 14 of the two chips together (make sure that the chips are oriented the same direction!).

5. Take a short length of wire-wrap wire, strip both ends to a length of about 1/4 inch. Tack solder one end to pin 6 of the 74LS86, following the cautions described above. Solder the other end into the solder pad previously occupied by pin 6 of the 74LS00. Make sure that you don't create any solder bridges when you do this.

6. Next, take another short length of stripped wire-wrap wire and connect one end it to pin 1 of the 74LS74, which you lifted its solder pad in step 3. The other end of this wire goes to the nearest end of the C82 (immediately to the left of the IC), as shown in the illustration in Figure 2.

That's it! A few cuts and a few solder tacks and we're done. A professional wouldn't take more than 30 minutes making this modification, and most of that would be dis-assembling and assembling the computer.

Reconnect all cables, making sure

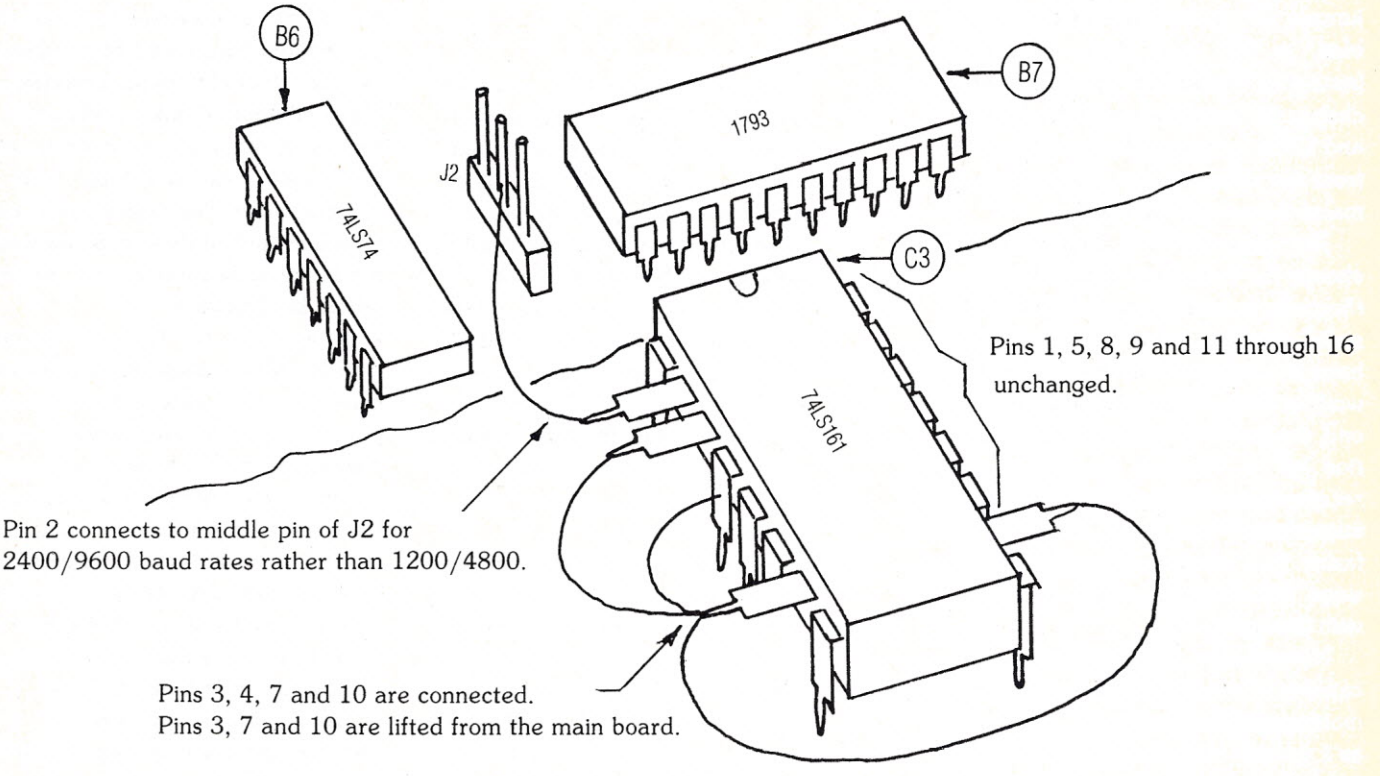


Figure 3. *Modifying the LS161 or LS163 IC at location C3 on the Osborne 1 main electronics board, to get a combination of faster baud rates for the serial port.*

that you properly align pin 1 (offsetting some of the cables can result in a very dead electronics board) and re-assemble the computer. When you turn the computer on, the cursor should be a nice solid white block.

As an interesting side benefit, changing the cursor with the above modification also results in underlined characters now being displayed in reverse video. The SuperCalc prompt, for example, becomes a solid reverse video bar instead of a thin underline.

If you used the public domain "blinking cursor program" (available from Osborne user groups) on your version of WordStar, you'll now have a blinking solid block. However, even without this software patch, it should be easy for you to find the cursor.

Speeding up the Serial Port

Our second change involves the serial port on the Osborne 1 computer. This change appears on page 21 and 22 of the Osborne 1 Technical Manual. Mike Iannamico and I, co-authors of the manual, managed to sneak this modification past the "engineering-powers-that-

were," but it is buried at the end of the Interface Design chapter, and many people who've read the manual didn't even notice it.

The Osborne 1 is capable of communicating at two baud rates. Normally, this is 300 and 1200 baud, the most commonly-used communication rates. By opening the computer and removing jumper J1, these baud rates are doubled, to 600 and 2400, respectively (note that the SETUP program used to change the baud rates still displays 300 and 1200 — jumper J cannot be sensed from the software, thus the message remains the same in SETUP).

The upper limit of 2400 baud was chosen for a number of reasons. First, the parts purchased by Osborne for the serial interface were only specified to have to work up to this rate (almost all work at far greater rates, however). Second, the fastest an Osborne 1 could function as a terminal (i.e., displaying received information and sending typed information) was 2400 baud — any faster and characters would be lost, since the software could not keep up with the hardware.

Nevertheless, the Osborne 1 is quite capable of receiving and sending information at higher baud rates, just as long as it doesn't have to display it or save it to disk while doing so. Four possible baud rate choices exist, two of which involve modifications:

300/1200	As the Osborne 1 comes.
600/2400	Remove jumper J1 inside the Osborne.
1200/4800	Basic modification described below.
2400/9600	basic modification plus additional wire.

Again, we'll be dealing with an IC soldered into the main electronics board of the Osborne 1 — in this case, UC3, a LS161 or LS163 at location C3. The Osborne 1 Technical Manual suggests that you de-solder this chip and replace it with a 16-pin socket. This is preferable to simply lifting and bending pins, especially if you may want to switch back and forth between pairs of baud rates at a future date.

Here's how to perform the baud rate modification (see figure 3):

1. Pins 2, 3, 7, and 10 of the 74LS161 (or 74LS163) must be lifted off the solder pads they are currently connected to. You may de-solder the chip, install a socket, bend these four pins up on the 74LS161 and re-insert it into the socket, or you may simply de-solder those four pins and bend them up.

2. Using wire-wrap wire, connect pins 3, 4, 7, and 10 together. Make sure only these pins are connected — solder bridges to other pins can cause problems. At this point, you have 1200/4800 baud instead of 300/1200 baud — where the SETUP program refers to the 300 baud, you get 1200, where SETUP says 1200, you get 4800. Jumper J1 may be in or out — it won't affect the transmission speed with this modification completed.

3. If you want the fastest baud rate combination, connect a wire from pin 2 of the 74LS161 to the middle pin on jumper J2 (located between 74LS74 at B6 and the 1793 at B7). You now have 2400 baud in place of 300 baud, and 9600 baud in place of 1200 baud.

That's all there is to it. Any cable that works correctly at the slower baud rates (i.e., handles the hardware handshake protocol of the serial port without losing characters) will work at these new, faster rates. The XON/XOFF protocol should still function correctly. It's as if you had given your Osborne 1 a shot of high octane gas — transmissions and receptions move faster.

One Last Caution

Toward the end of Osborne 1 production history Osborne started receiving a new, multi-layered main logic board. I was unable to ascertain, from my contacts at the company, when and how many of these multi-layered boards were put into production. Soldering directly on a multi-layer board can be quite tricky, because you may get the connection on the surface, but fail to make contact underneath. If you have a late model Osborne 1 and have problems with the modification, the multi-layered board may be the culprit — have a professional repair technician look at your work and check it out.

These two changes are by no means all the modifications that can be performed on an Osborne 1. Osborne experimented by created prototypes with 80-track drives (368k per diskette), double-sided drives (736k per diskette), and had even replaced the B drive with a half-height winchester-type hard disk and controller. The primary problem involved in each of these changes was not hardware (slide out one drive, slip in another), but software.

In addition, the Osborne Executive logic board was originally designed to fit in the Osborne 1 case and provide an upgrade to 80 columns (with a few other niceties thrown in). A seven-inch monitor version of the Osborne 1 was discussed, but rejected in favor of the Executive 1. Indeed, a whole new computer, code-named Vixen, was designed. It had everything a fully equipped Osborne 1 did (and more), but weighed in at 18 pounds and measured only half the size.

On the purely software side, I prepared a 52-column version of WordStar in mid-1982, but the then-promised release by MicroPro of a new version of WordStar cancelled the release of the 52-column version. The UCSD p-system was worked up for the Osborne 1 double-density option, but again, new version releases pushed back the introduction date until it was too late. A new HELP utility was also put together, as were a few other simple upgraded offerings, but most of these never saw the dealer's shelves, either.

(Fortunately, the plan to re-organize Osborne Computer Corporation is up for final approval. If and when that happens, we will report the latest product offerings — editors).

It is my hope in offering these two modifications that Osborne 1 computers will continue to be put to good use. Perhaps, when all the dust has settled, some of these modifications will become part of service agreements or new products. We Osborne 1 owners can only hope.

Rescue From Toxic Spills

The program GAUSS calculates a Gaussian dispersion model of gases released from a toxic spill. It runs on an Osborne 1 or Executive under MBASIC.

William White

A tank truck carrying a toxic chemical is traveling along a rural road near a south California community. In the early Sunday morning hours, a light fog hangs over the valley making the winding road treacherous. A passenger car in the opposite lane has a blowout. Car and truck collide. Tires screech, metal crunches, radiators hiss, and then almost silence. The driver of the car lies unconscious on the roadway, his vehicle wedged under the tank truck, now punctured and leaking. The truck driver scrambles from the cab, stunned but apparently uninjured. The truck driver quickly calls for police and fire assistance. Upon questioning by the police dispatcher, the driver reads the shipping manifest. He informs the dispatcher that the chemical he is hauling is anhydrous ammonia.

The dispatcher calls CHEMTREC (Chemical Transportation Emergency Center) for instructions on how to handle the chemical spill. After providing brief directions on handling the spill, the CHEMTREC duty officer calls the truck firm and the shipper. Within minutes your phone rings waking you from an enjoyable dream. As Transportation Emergency Team Leader for your company, you are requested to go to the scene and render assistance. You phone your plant and request that other team members be dispatched with the vehicle containing various patching devices.

On the way to the scene, many thoughts race through your head. How bad is the leak? Will you be able to stop it with the emergency capping kit? Has anyone been exposed to the chemical? Have people been evacuated from the downwind areas? Will the news media

As you drive back home to your sleeping family, you realize that your fast response and accurate information to the fire chief would not have been possible without your Osborne.

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